INCH-POUND
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PERFORMANCE SPECIFICATION

CLOTH, DUCK, COTTON, PLIED YARNS, FIRE, WATER, WEATHER AND MILDEW RESISTANT TREATED, LIGHTDRY FINISH

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers one (1) type of finished cloth duck, cotton, plied yarns, treated for fire, water, weather and mildew resistance, using a light dry finish.

1.2 Classification. Not applicable

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textiles Materials

(Copies of these (this) documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 - Rules and Regulations Under the Textile Fiber Product Identification Act
Title 16 CFR Part 1500 - Federal Hazardous Substances Act Regulations
Title 40 CFR - Protection of Environment

(Copies of these documents are available online at https://www.ecfr.gov.)

ENVIRONMENTAL PROTECTION AGENCY

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
(40 CFR Part 162) State Registration of Pesticide Products

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
FIFRA as amended by the Food Quality Protection Act of 1996 and the
Pesticide Registration Improvement Act

(Copies of these (this) documents are available online at https://www.epa.gov/pesticides.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM30 - Test Method for Antifungal Activity, Assessment on Textile Material: Mildew and Rot Resistance of Textile Materials
AATCC TM169 - Test Method for Weather Resistance of Textiles: Xenon Lamp Exposure

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D751 - Standard Test Method for Coated Fabrics
ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D1776/D1776M - Standard Test Method for Conditioning and Testing Textiles
ASTM D3776/D3776M - Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6413/D6413M - Standard Test Methods for Flame Resistance of Textiles (Vertical Test)

(Copies of these documents are available online at https://www.astm.org.)

OTHER PUBLICATIONS

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Replica set is available for purchase from SDL Atlas Customer Service, 1-803-329-2110.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.1.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Figure. Figure 1 is furnished for informational purposes only. To the extent of any inconsistencies between the written document and the figure, the written document shall govern.

3.4 Materials.

3.4.1 Yarn. The warp and filling yarns shall be made from cotton fiber which has been carded, drawn, spun, and twisted into 2-ply yarns when tested as specified in 4.5.

3.4.2 Weave. The weave shall be plain. The cloth shall be woven on either a fly-shuttle or shuttleless loom. When the cloth is woven on shuttleless looms, the fringed selvage shall be secured by a leno-lock stitch and the protruding fringe shall not exceed 3/16-inch.

3.5 Color. The color of the finished cloth shall be as specified in the contract or procurement documents (see 6.2) and shall match the standard sample.

3.5.1 Visual shade matching. The color and appearance of the dyed cloth shall match the standard sample when tested as specified in 4.5.

3.5.2 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be “Free” of labile sulfur when tested as specified in 4.5.

3.5.3 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished cloth referenced in 3.4 shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

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TABLE I. Colorfastness requirements.

Color Evaluation	Weathering (after 100 kJ/m^2), grade, (min.)	Crocking (Dry/Wet), grade, (min.)
All colors (Olive Green 107, Olive Drab 7, Desert Tan 459, Tan 380, and Tan 686)	3	1.5

3.6 Spectral reflectance.

3.6.1 Solid colors. The reflectance values shall conform to the requirements listed below, in Table II, when tested as specified in 4.5.

TABLE II. Spectral reflectance requirements.

Reflectance values (percent)				
Wavelength, Nanometers (nm)	Olive Green 107 and Olive Drab 7		Desert Tan 459, Tan 380 and Tan 686	
	Min.	Max.	Min.	Max.
600	3	12	--	--
620	3	12	--	--
640	3	12	--	--
660	3	13	--	--
680	3	15	--	--
700	3	21	25	58
720	5	30	25	59
740	7	42	25	61
760	11	50	26	62
780	17	56	27	63
800	24	60	28	64
820	32	63	30	66
840	37	65	33	68
860	40	66	36	69

3.7 Physical requirements. The finished cloth shall conform to the requirements, listed below, in Table III, when tested as specified in 4.5.

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TABLE III. Physical requirements.

Characteristic	Requirement	
	Greige	Finished
Weight, oz./sq.yd. (min.)	10	17
Breaking strength, pounds (min.)		
Warp	170	185
Filling	150	165
Tearing strength, pounds (min.)		
Warp	6.0	--
Filling	6.0	--
Air permeability, cu.ft./min./sq.ft., (max.)	4.0	--

3.8 Finish.

3.8.1 Water repellency. The cloth shall be given a durable water repellent treatment. The cloth shall meet the requirements in Table IV, when tested as specified in 4.5.

TABLE IV. Water repellency.

	Hydrostatic resistance, ml (max.)
Initial	50
After weathering	50

3.8.2 Flame resistance. The finished cloth shall meet the requirements in Table V, when tested as specified in 4.5.

TABLE V. Flame resistance.

	Flame resistance	
	After flame (sec.), (max.)	Average char length (in.), (max.)
Initial	2.0	4.5
After weathering	2.0	4.5
After water leaching	2.0	4.5

3.8.3 Mildew resistance. The finished cloth shall be treated with an EPA approved organic mildew inhibitor and shall retain 75 percent breaking strength after 12 weeks of exposure, initially and after leaching when tested as specified in 4.5.

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3.8.4 Flexibility. The finished cloth shall show a bending moment of not more than 0.013 inch-pound initially and not more than 0.032 in-pound after exposure at 0° F (± 5) when tested as specified in 4.6.6.

3.8.5 Dimensional stability. The shrinkage or elongation both in the warp and filling of the finished cloth shall not be greater than 2.0 percent in either direction, when tested as specified in 4.5.5

3.9 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.5.

3.10 Face/back identification. The warp side shall be identified as the face side by stamping that side with the word “FACE” or “Back” at each end of the roll. The marking shall be clearly legible with any indelible marking used commercially.

3.11 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width exclusive of the selvage.

3.12 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.13 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1)
b. Conformance inspection (see 4.3.2)

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

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4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for appearance, color, the defects of 4.4 and tested for the characteristics as specified in 4.5 through 4.7.

4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

4.4 Visual Examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Table VI and Type I of MIL-STD-3064 except for slubs and knots which shall be in accordance with Table VI.

TABLE VI. Defects descriptions.

Creases in the fabric
Wrong draw, appearing as a line in the warp direction for more than 18" in length
Slubs greater than 1-1/2" long and greater than 1/16" wide
Knots 1/8" or greater

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The finished cloth shall be tested for the characteristics listed in Table VII. The methods of testing as specified wherever applicable and as listed in Table VII shall be followed. The physical and chemical values specified in section 3, (except where otherwise indicated) apply to the results of determinations made on the sample unit for test purposes as specified in the applicable test methods, except that for the finished cloth weight, the lot average shall apply. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 2 (two) continuous yard full width of greige cloth and four (4) continuous yard full width of the finished cloth for all physical and chemical tests. For all requirements except weight, the lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. For the weight requirement, the lot shall be unacceptable if the average of the results of all determinations fails to meet any test requirement specified. The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VII. Material testing.

Characteristic	Requirement Reference	Test method
Fiber identification	See 3.4.1	AATCC TM20 (see 6.4)
Yarn ply	See 3.4.1	Visual
Weave	See 3.4.2	Visual
Visual shade matching	See 3.5.1	See 4.6.1
Labile sulfur	See 3.5.2	See 4.6.2
Colorfastness: Weathering Crocking	 See Table I See Table I	 AATCC TM169 and see 4.6.3, 1/ AATCC TM8, 2/
Spectral reflectance	See Table II	See 4.6.4

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TABLE VII. Material testing - Continued

Characteristic	Requirement Reference	Test method
Weight: Greige cloth Finished cloth	 See Table III See Table III	 ASTM D3776/D3776M ASTM D3776/D3776M
Breaking strength: Geige cloth Finished cloth	 See Table III See Table III	 ASTM D5034 ASTM D5034
Tearing strength: Geige cloth Finished cloth	 See Table III See Table III	 ASTM D1424 ASTM D1424
Air permeability	See Table III	ASTM D737
Water repellency: Initial After weathering	 See Table IV See Table IV	 ASTM D751, Procedure B, Procedure 2, 3/ AATCC TM169 and ASTM D751, Procedure B and Procedure 2
Flame resistance: Initial After weathering After water leaching	 See Table V See Table V See Table V	 ASTM D6413/D6413M 4/ AATCC TM169 and ASTM 6413/D6413M, 4/, 5/ See 4.6.4 and ASTM D6413/D6413M, 4/, 6/
Mildew resistance Initial After leaching	See 3.8.3	 AATCC TM30, Test I, 7/, 8/ See 4.6.5, AATCC TM30, Test I, 7/
Flexibility Initial At low temperature	See 3.8.4	 See 4.6.6 See 4.6.6.1
Dimensional stability	See 3.8.5	See 4.6.7
Toxicity	See 3.8.6	See 4.7

1/ Rated using the AATCC EP1.
2/ Rated using the AATCC EP8
3/ The exposure time shall be 10 minutes and the hydrostatic head shall be 10-inches. If the test result fails to comply with 3.8.1, the test specimens shall be immersed in water at a temperature of 75 – 85°F to a depth of 2-inches for two (2) hours and then retested while still wet. If the retested specimens meet requirements of 3.8.1, the cloth shall be considered as having passed the water resistance test. For accelerated weathering procedure the specimen size shall be 6 by 6-inches.
4/ Four (4) specimens in each direction on (warp and filling) shall be tested.
5/ The specimens shall be selected from samples subjected to the accelerated weathering procedure specified in 4.6.3.

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6/ The specimen shall be leached for 24 hours before being tested for flame resistance after water leaching.
7/ A minimum of 5 specimens shall be tested.
8/ The contractor shall indicate on his test report which fungicide was used.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 7.4.1) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the cloth shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.5.1). The cloth shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

4.6.2 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g SnCl2 (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin (II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin (II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.

4.6.3 Accelerated weathering test. The test procedure shall be in accordance with AATCC TM169 with the following exceptions:

a. The weathering test cycle shall be 40 minutes of light, 20 minutes of light with water spray on the fabric face, 60 minutes of light, 60 minutes of darkness. The test cycle shall be repeated until the total energy exposure is equal to 100 kilojoules per square meter (kJ/m^2).

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b. The irradiance level shall be 0.55 (± 0.01) watt/square meter/nanometer ($W/m^2/nm$) bandpass at 340 nanometers.

c. The relative humidity shall be 50 (± 5) percent during the light cycle and not lower than 95 percent during the dark cycle.

d. The control set points shall be as follows:

	Dark cycle	Light cycle
Black panel	100 ºF	170 ºF
Conditioning water	104 ºF	127 ºF
Wet bulb depression 1/	32 ºF	50 ºF

1/ As a guide only: Adjust to achieve required relative humidity (see c. above)

e. The test specimens shall fit the specimen rack of the apparatus with no wrinkles or gaps. The test specimen shall be mounted on the outside of the rack with the use of appropriate stainless steel spring clips or equivalent.

4.6.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for solid colors at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with two (2) layers of the same fabric for all colors.

The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns/ wales or courses when presented to the sample port. Camouflage materials should be measured with the appropriate aperture size to ensure that only one (1) color is measured at a time. Always try to use the largest aperture possible. Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.6.5 Leaching resistance. Specimens to be tested for mildew and rot resistance shall be submerged in a suitable tank containing water at a temperature of 80 to 85ºF and allowed to remain immersed for a period of 24 hours. Water flow through a tube into the tank is adjusted to

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ensure a completed change of water not less than three (3) times in 24 hours. The pH and temperature of the water is recorded and included in the report of the results.

4.6.6 Flexibility test. Flexibility testing shall be conducted in accordance with the following procedure:

1. Test specimen. Unless otherwise specified in the procurement document, the specimen shall be a rectangle of cloth with the long dimension parallel to the yarns to be tested. The specimen shall be cut from the smoothest area possible which has not been previously folded or in any manner deformed. Eight (8) specimens, 1 inch by 2 inches, shall be cut, four (4) with the long dimension in the warp direction and four (4) in the filling direction, and pressed between two (2) 6 by 6-inch, glass plates weighted with a 20-pound weight for four (4) hours at standard conditions as specified in ASTM D1776/D1776M, prior to testing.

2. Apparatus. The apparatus for determining the stiffness of cloth, shall be in accordance with Figure 1. It shall be a cantilever beam bending type, consisting essentially of the following:

2.1 Specimen vise. Specimen vise, "V", at least one (1) inch wide, to which the pointer indicator "I_2" is attached, and which can be rotated in a clockwise direction alternately by means of a hand crank or motor about the point O, at a rate of 60 degrees per minute.

2.2 Pendulum weighing system. Pendulum weighing system, including an angular defection scale pointer indicator "I_1", adjustable bending plate "Q" for contacting the free end of the specimen, and a series of detachable weights "M" calibrated to the pendulum system. This system shall be pivoted for nearly frictionless rotation about the point O.

2.3 Angular deflection scale. Angular deflection scale calibrated in degrees of arc which will indicate the difference in the angles through which the vise has been turned and the load of the pendulum has been deflected.

2.4 Fixed load scale. Fixed load scale which measures the deflection O of the pendulum system, calibrated to directly read 100 L sin Θ, where "L" is the distance between the center of rotation O, and the center of the applied load "M". Thus "M" times the load scale reading divided by 100 gives the bending moment directly.

3. Procedure. Testing shall be conducted under standard conditions (see 1.). The machine shall be placed in test position and leveled. A suitable moment weight shall be placed on the pendulum and then, if necessary, the load scale adjusted to indicate zero. The moment weight applicable to a given cloth may be determined by trial. The bending plate shall be set to a 1/2 -inch span between the edge of the vise and the nearest edge of the bending plate. The motor shall be started and kept running throughout the test since its vibration minimizes friction effects in the weighting system. However, during the initial portion of the test, it shall be disengaged.

3.1 The specimen shall be clamped firmly in the vise and the long edge parallel to the face of the dial plate. Care shall be taken that the specimen is not unduly deformed in handling.

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The face surface of the specimen (weave face, finished side, coated side, etc.) shall be placed upward in the vise.

3.2 Sufficient load shall be applied to the specimen by turning the hand crank to show a 1-percent load reading. The angle pointer shall then be set to zero (0) and subsequent load readings reduced by one (1) division.

3.3 The motor engaging lever shall be held down and the load scale readings taken at 20-degree angular deflection of the specimen.

4. Calculation of results. The stiffness shall be calculated as follows:

$$\text{Bending moment, inch-pound} = \frac{\text{load scale reading X moment weight}}{100}$$

4.1 Determinations. Average of four determinations and reported to the nearest 0.001 inch-pound in each direction.

4.6.6.1 Flexibility at low temperature. Flexibility test specimen shall be prepared as specified in 4.6.6. The weight shall then be removed. The plate/specimen assembly and the test instrument shall then be subjected to a temperature of 0°F (±) 5 for not less than one (1) hour and then tested at that temperature. Four (4) specimens shall be tested in each direction (warp and filling) and their results averaged respectively and reported to the nearest 0.001 inch-pound in each direction.

4.6.7 Dimensional stability. A length of cloth approximately 40 inches long shall be drawn from the finished roll (the cloth may be left intact), and immediately marked by two (2) lines three (3) feet apart and perpendicular to the selvage. This sample length shall be then lay on a flat horizontal surface under standard conditions as specified in ASTM D1776/D1776M for at least 24 hours. The distance between the lines marking the 3-foot distance shall be re-measured at three locations and the average percent change in length for the three (3) locations recorded.

4.7 Toxicity test. When the toxicity requirement (see 3.16) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.4 and 6.6).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cotton duck cloth treated for fire, water, weather, and mildew resistance is intended for use in the manufacturing of tents and tent flys.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.1).
c. When first article is required (see 3.1).
d. Color required (see 3.5).
e. Water repellency requirement, if other than specified (see 3.8.1).
f. Mildew inhibitor, if other than specified (see 3.8.3).
g. Width of cloth required (see 3.11).
h. Length required if other than specified (see 3.12).
i. Inspection conditions (see 4.2).
j. Conformance inspection acceptance quality limits (AQLs) (see 4.3.2).
k. Packaging (see 5.1).

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35°F (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens".

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6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.6 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1)

Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250).

6.7 Subject term (key word) listing.

Tent
Tent flys

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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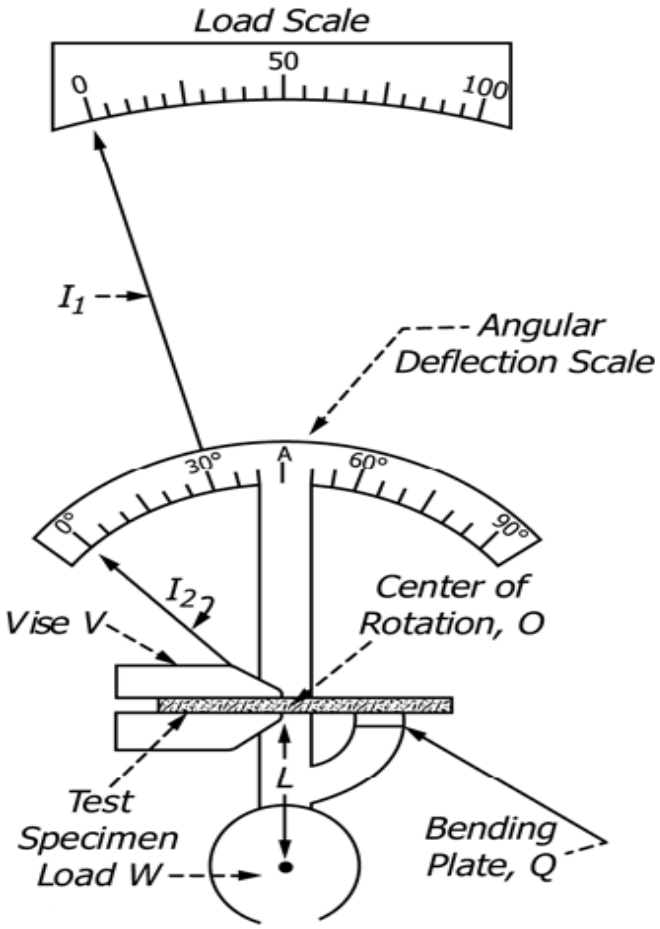


FIGURE 1. Test Apparatus.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA-CT

(Project 8305-2023-006)

Review activities:
Army - MD
Navy – MC

NOTE: The activities listed above were interested in this document as of date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST online database at https://assist.dla.mil.

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